Ref: HRSU/SDS/CNT/001 **Revision:** 01 **Date:** 14 November 2026

SAFETY DATA SHEET: CALCIUM NITRATE TETRAHYDRATE (INDUSTRIAL GRADE)

Prepared in the 16-section GHS format. Supersedes the earlier undated issue.

SECTION 1: PRODUCT AND COMPANY IDENTIFICATION

- **Product Name:** Calcium Nitrate Tetrahydrate – Industrial Grade (Crystalline Powder)
- **Other Names:** Nitric acid, calcium salt, tetrahydrate; Norwegian saltpeter tetrahydrate; Lime nitrate; Calcium dinitrate tetrahydrate
- **CAS No.:** 13477-34-4
- **EC (EINECS) Number:** 233-332-1 (calcium nitrate, REACH-registered; hydrate-specific EC: 603-865-8)
- **Molecular Formula:** $Ca(NO_3)_2 \cdot 4H_2O$ **Molecular Weight:** 236.15 g/mol **HSN:** 28342990
- **Relevant Uses:** Industrial use only (wastewater and odour control, concrete admixtures, latex/rubber coagulant, mining and oilfield uses). Not for food, feed or pharmaceutical use.
- **Supplier:**
  - **Entity Name:** HRSU Indore Pvt. Ltd.
  - **Registered Office:** UG-3, Abhishek Avenue, Slice-4, Scheme No. 78, Vijay Nagar, Indore (M.P.), India, 452010
  - **Manufacturing Unit:** Plot No. 62 (old 53), MPIDC Industrial Area, Maksi, District Shajapur (M.P.), India, 465106
  - **Email:** info@hrsuindore.com **Website:** hrsuindore.com
  - **Telephone / Emergency Contact:** +91 9425000484

SECTION 2: HAZARDS IDENTIFICATION

- **GHS Classification:**
  - Oxidizing solids: Category 3
  - Acute toxicity, oral: Category 4
  - Serious eye damage: Category 1
- **GHS Label Elements:**
  - **Pictograms:**



  - **Signal Word: Danger**
  - **Hazard Statements:**
    - H272: May intensify fire; oxidizer.
    - H302: Harmful if swallowed.
    - H318: Causes serious eye damage.
  - **Precautionary Statements:**
    - P210: Keep away from heat, sparks, open flames and hot surfaces. No smoking.
    - P220: Keep away from clothing and other combustible materials.

HRSU Indore Pvt. Ltd.
Website: www.hrsuindore.com
Email: info@hrsuindore.com
Contact: +91 9425000484
GSTIN: 23AAGCH4199C1ZP
Registered Office: UG - 3, Abhishek Avenue, Slice - 4, Scheme. No. 78, Vijay Nagar, Indore (M.P.), 452010
Manufacturing Unit: Plot No. 62 (Old 53) Mpidc, Industrial Area Maksi District - Shajapur (M.P.), 465106

- P264: Wash hands thoroughly after handling.
- P270: Do not eat, drink or smoke when using this product.
- P280: Wear protective gloves, protective clothing and eye/face protection.
- P301+P312: IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell.
- P305+P351+P338: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
- P310: Immediately call a POISON CENTER or doctor.
- P370+P378: In case of fire: Use water spray to extinguish.
- P501: Dispose of contents/container in accordance with local regulations through an authorized agent.

- **Other Hazards:** Contact with combustible, organic or reducing materials may cause fire. Dust may irritate the respiratory tract. Hygroscopic.

SECTION 3: COMPOSITION / INFORMATION ON INGREDIENTS

Substance	CAS No.	EC No.	Content	Classification
Calcium nitrate tetrahydrate	13477-34-4	233-332-1 / 603-865-8	98.0% min (assay as $Ca(NO_3)_2 \cdot 4H_2O$)	Ox. Sol. 3 (H272); Acute Tox. 4 (H302); Eye Dam. 1 (H318)

Remainder: free moisture (max 2.0%) and water-insoluble matter (max 0.15%), per product specification.

SECTION 4: FIRST AID MEASURES

- **Inhalation:** Remove to fresh air. If breathing is difficult, give oxygen; if not breathing, give artificial respiration. Get medical attention.
- **Ingestion:** Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Call a poison center or doctor immediately.
- **Skin Contact:** Immediately flush skin with plenty of water for at least 15 minutes. Remove contaminated clothing. Get medical attention if irritation persists.
- **Eye Contact:** Immediately flush eyes with plenty of water for at least 15 minutes, lifting upper and lower eyelids occasionally. Remove contact lenses if easy to do. Get immediate medical attention (risk of serious eye damage).
- **Most Important Symptoms:** Severe eye irritation/damage; irritation of the mouth, throat and stomach if swallowed. Large ingestion of nitrates may cause methaemoglobinaemia (bluish skin, breathlessness).

SECTION 5: FIRE-FIGHTING MEASURES

- **Flammability:** Non-combustible, but as an oxidizer it will support and intensify combustion of other materials.
- **Extinguishing Media:** Use flooding quantities of water or water spray, and any means suitable for the surrounding fire.
- **Special Hazards:** May accelerate burning. Heating or fire releases toxic nitrogen oxides. Containers may explode when heated in a fire.
- **Protective Equipment:** Wear full protective clothing and self-contained breathing apparatus.

SECTION 6: ACCIDENTAL RELEASE MEASURES

- **Personal Precautions:** Ventilate the area. Avoid breathing dust. Wear gloves, chemical goggles and protective clothing.
- **Environmental Precautions:** Do not let the product enter drains or water courses.
- **Cleanup Procedures:** Keep away from combustible materials (wood, paper, oil, sawdust). Sweep or scoop spilled solid into a clean, dry, labelled container for disposal; do not return spilled product to stock. Wash the contaminated surface with water and

HRSU Indore Pvt. Ltd.
Website: www.hrsuindore.com
Email: info@hrsuindore.com
Contact: +91 9425000484
GSTIN: 23AAGCH4199C1ZP
Registered Office: UG - 3, Abhishek Avenue, Slice - 4, Scheme. No. 78, Vijay Nagar, Indore (M.P.), 452010
Manufacturing Unit: Plot No. 62 (Old 53) Mpidc, Industrial Area Maksi District - Shajapur (M.P.), 465106

dispose of according to local authority requirements.

SECTION 7: HANDLING AND STORAGE

- **Handling:** Minimize dust generation. Avoid contact with skin, eyes and clothing. Keep away from heat and ignition sources, combustible and organic materials. Wash thoroughly after handling.
- **Storage:** Store in a cool, dry, well-ventilated area, away from heat, sunlight and moisture, in tightly closed original packaging (product is hygroscopic). Store separately from combustible materials, organic materials, reducing agents, ammonium salts, strong acids and powdered metals.

SECTION 8: EXPOSURE CONTROLS / PERSONAL PROTECTION

- **Exposure Limits:** No specific occupational exposure limit established; keep dust below applicable nuisance-dust limits.
- **Ventilation:** Local and/or general exhaust ventilation is recommended.
- **Respiratory Protection:** Where dust is apparent, use a particulate respirator (P2/N95 or better).
- **Skin Protection:** Wear protective gloves and clean body-covering clothing.
- **Eye Protection:** Wear chemical safety goggles or a full-face shield (risk of serious eye damage). Provide an eye-wash station near the work area.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

- **Appearance:** Colourless to white crystalline powder (hygroscopic)
- **Odor:** Odorless
- **pH:** 4.5 – 7.0 (5% w/v aqueous solution)
- **Melting Point:** approx. 43 °C (melts in its water of crystallization)
- **Solubility:** Freely soluble in water
- **Molecular Weight:** 236.15 g/mol
- **Relative Density:** approx. 1.82 (literature value)
- **Flammability:** Non-combustible; oxidizing solid

SECTION 10: STABILITY AND REACTIVITY

- **Reactivity / Stability:** Stable under ordinary conditions of use and storage. Strong oxidizer.
- **Conditions to Avoid:** Heat, moisture, and contamination with combustible or organic materials.
- **Incompatibility:** Reducing agents, organic and combustible materials, ammonium salts, strong acids, powdered metals.
- **Hazardous Decomposition Products:** Toxic fumes (nitrogen oxides, calcium oxide) when heated to decomposition.

SECTION 11: TOXICOLOGICAL INFORMATION

- **Acute Oral Toxicity:** Classified Category 4 (H302), consistent with major supplier classifications. Reported rat oral LD50 values in the literature range from approx. 300 to 3,900 mg/kg.
- **Eye:** Causes serious eye damage (Category 1).
- **Skin:** May cause mild skin irritation; not classified.
- **Inhalation:** Dust may irritate the nose, throat and respiratory tract.
- **Carcinogenic Effects:** Not identified as a carcinogen by IARC.

SECTION 12: ECOLOGICAL INFORMATION

- **Toxicity to Fish:** LC50 – *Oncorhynchus mykiss* (rainbow trout): > 98.9 mg/L, 96 h.

HRSU Indore Pvt. Ltd.
Website: www.hrsuindore.com
Email: info@hrsuindore.com
Contact: +91 9425000484
GSTIN: 23AAGCH4199C1ZP
Registered Office: UG - 3, Abhishek Avenue, Slice - 4, Scheme. No. 78, Vijay Nagar, Indore (M.P.), 452010
Manufacturing Unit: Plot No. 62 (Old 53) Mpidc, Industrial Area Maksi District - Shajapur (M.P.), 465106

- **PBT and vPvB Assessment:** Not considered persistent, bioaccumulative or toxic (PBT).
- **Other:** Large releases of nitrate to water bodies can contribute to eutrophication. Do not discharge into drains or waterways.

SECTION 13: DISPOSAL CONSIDERATIONS

- Dispose of unused product and contaminated material through an appropriate, approved waste disposal facility in accordance with national, state and local regulations.
- Empty bags may retain product residue: handle as the product and do not reuse them for other materials.

SECTION 14: TRANSPORT INFORMATION

Regulation	UN No.	Proper Shipping Name	Class	Packing Group
IMDG (Sea)	1454	CALCIUM NITRATE	5.1 (Oxidizing substances)	III
IATA / ICAO (Air)	1454	Calcium nitrate	5.1	III
ADR / RID (Road / Rail)	1454	CALCIUM NITRATE	5.1	III
DOT (USA) / TDG (Canada)	1454	Calcium Nitrate	5.1	III

- **Label:** Class 5.1 (Oxidizer). **Subsidiary Hazard:** None. **Marine Pollutant:** No.
- **Packaging:** 25 kg / 50 kg polypropylene (PP) woven bags with inner polyethylene (PE) liner; UN-certified packaging to be used for export by sea or air.
- **Special Precautions:** Segregate from combustible materials, reducing agents, ammonium salts and acids as required for Class 5.1 under the applicable transport code.

SECTION 15: REGULATORY INFORMATION

- **Transport Classification:** UN 1454, Class 5.1, PG III (see Section 14).
- **USA:**
  - SARA 302: No chemicals in this material are subject to the reporting requirements of SARA Title III, Section 302.
  - SARA 311/312 Hazards: Reactivity Hazard.
  - SARA 313: This material contains calcium nitrate tetrahydrate, which is subject to reporting.
  - California Prop. 65: This product does not contain any chemicals known to the State of California to cause cancer or reproductive harm.

SECTION 16: OTHER INFORMATION

- **Revision 01 (14 November 2026):** GHS classification (Sections 2, 3, 11) and transport information (Section 14, IMDG/IATA) aligned with UN 1454 / Class 5.1 / PG III; first aid, fire-fighting, handling and protection sections expanded; company address aligned with registered office.
- **Disclaimer:** The information provided is believed to be accurate and represents the best information currently available. However, we make no warranty of merchantability or any other warranty, express or implied, with respect to such information.

HRSU Indore Pvt. Ltd.
Website: www.hrsuindore.com
Email: info@hrsuindore.com
Contact: +91 9425000484
GSTIN: 23AAGCH4199C1ZP
Registered Office: UG - 3, Abhishek Avenue, Slice - 4, Scheme. No. 78, Vijay Nagar, Indore (M.P.), 452010
Manufacturing Unit: Plot No. 62 (Old 53) Mpidc, Industrial Area Maksi District - Shajapur (M.P.), 465106